Module 2: Relationships Between Variables

Lesson 4 | Compare proportions and describe association

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What we will learn

Lessons 1–3 describe variables individually and compare numerical distributions across groups. We now ask how **two variables relate within the same observations**.

By the end, you should be able to:

- Construct a crosstab and distinguish joint, marginal, and conditional percentages.
- Choose a denominator that answers the stated question.
- Read a scatterplot for direction, form, strength, and unusual observations.
- Calculate sample covariance and Pearson correlation, with their units and limitations.
- Describe association without claiming causation.

Reading: main 8th-edition textbook, §2.3, pp.65–69 (crosstabs and percentages), §2.4, pp.75–80 (scatterplots and comparative charts), and §3.5, pp.144–152 (covariance and correlation). These are selected descriptive topics, rather than every procedure in those sections.

Choose tools for the variable pair

Variable pair	Useful display	Numerical summary
Two categorical variables	Crosstab, grouped bars, or percentage bars	Counts and conditional percentages
Categorical and numerical	Side-by-side boxplots or aligned distribution plots	Group sizes, centers, and spreads

Two numerical variables	Scatterplot	Covariance and correlation when appropriate

A numerical variable can be grouped into classes for a crosstab. This simplifies comparisons but loses within-class detail. Keep the original values for numerical analysis. The Restaurant boxplots in Lesson 3 and the price-band tables below answer related questions at different resolutions.

Business case: quality ratings and meal-price bands

An analyst studies **300 Los Angeles restaurants** from the book's historical teaching dataset. Each restaurant has an ordered quality rating and a meal price in US dollars. The analyst asks: **Within each quality category, how are restaurants distributed across price bands?** A second question reverses the conditioning: **Within a price band, how are quality ratings distributed?** These questions use the same records but different denominators.

يدرس محلّل بيانات **ثلاثمئة مطعم في لوس أنجلوس** من بيانات الكتاب التعليمية التاريخية. لكل مطعم تقييم جودة مرتب وسعر وجبة بالدولار الأمريكي. يسأل المحلّل: **داخل كل فئة من فئات الجودة، كيف تتوزع المطاعم على فئات الأسعار؟** ثم يعكس السؤال: **داخل فئة سعرية محددة، كيف تتوزع تقييمات الجودة؟** يستخدم السؤالان السجلات نفسها، لكن مقام حساب النسبة يختلف بينهما.

Download Restaurant CSV. One row is one restaurant. The four bands are **[10,20), [20,30), [30,40), [40,50)** USD: include the lower boundary and exclude the upper one. A price of USD 20 belongs to the second band. For these whole-dollar observations, these correspond to the book's USD 10–19, 20–29, 30–39, and 40–49 classes.

Crosstabs: count combinations

A **crosstabulation** counts observations classified by two variables. Each internal cell describes a combination. **Marginal totals** describe each variable separately.

```
# addmargins() adds row totals, column totals, and the grand total to the crosstab.
# kable() displays counts, not percentages; each internal cell counts one combination.
kable(addmargins(counts), caption = "Restaurants by quality and meal-price band (USD).
 ↪  Sum denotes totals.")
```

Table 1: Restaurants by quality and meal-price band (USD). Sum denotes totals.

	10–<20	20–<30	30–<40	40–<50	Sum
Good	42	40	2	0	84
Very Good	34	64	46	6	150
Excellent	2	14	28	22	66
Sum	78	118	76	28	300

The cell **64** represents Very Good restaurants priced from USD 20 up to, but below, USD 30. The Very Good row total is **150**, the second price-band column total is **118**, and the grand total is **300**. A restaurant contributes to exactly one internal cell.

Joint, marginal, and conditional percentages

The same count can answer different questions:

Question	Denominator	Calculation	Interpretation
What share of all restaurants are Very Good **and** in the second price band?	All 300 restaurants	$64/300 \times 100 = 21.33\%$	Joint percentage
What share of all restaurants are Very Good?	All 300 restaurants	$150/300 \times 100 = 50\%$	Marginal percentage
Among Very Good restaurants, what share are in the second price band?	The 150 Very Good restaurants	$64/150 \times 100 = 42.67\%$	Conditional row percentage
Among restaurants in the second price band, what share are Very Good?	The 118 restaurants in that band	$64/118 \times 100 = 54.24\%$	Conditional column percentage

Say "among whom?" before calculating a percentage. A percentage without its reference group can answer the wrong question even when the arithmetic is correct.

Row percentages: price distribution within quality

```
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# rowSums(): sums each row separately.
# row_pct was calculated with prop.table(counts, margin = 1): divide by each row total.
# cbind() adds the percentage totals; digits = 1 rounds only the displayed values.
kable(cbind(row_pct, Total = rowSums(row_pct)), digits = 1,
      caption = "Within each quality category: percentage in each price band. Each row
       ↪  totals 100% before rounding.")
```

Table 2: Within each quality category: percentage in each price band. Each row totals 100% before rounding.

	10–<20	20–<30	30–<40	40–<50	Total
Good	50.0	47.6	2.4	0.0	100
Very Good	22.7	42.7	30.7	4.0	100
Excellent	3.0	21.2	42.4	33.3	100

For Good restaurants, **50.0%** fall in the lowest band. For Excellent restaurants, **75.8%** cost at least USD 30: $(28 + 22)/66 \times 100$. The comparable share for Good is **2.4%**: $(2 + 0)/84 \times 100$. These distributions differ substantially in this sample.

The groups contain 84, 150, and 66 restaurants. Conditional percentages make their distributions comparable despite unequal group sizes. Counts should still accompany the comparison.

```
# factor(): stores categorical values; levels fixes their permitted order and labels can
 ↪  rename their display.
# ifelse(): chooses a value element by element according to a logical condition.
# sprintf(): formats text using placeholders such as %.2f for two decimal places.
# paste0(): joins text with no separator; collapse can combine a vector into one string.
# rowSums(): sums each row separately.
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# geom_col(): draws bars at supplied heights; it does not count records.
```

```r
# geom_text(): draws labels from mapped data; position and hjust/vjust control placement.
# scale_fill_manual(): assigns explicit fill colors to categories; named values preserve
 ↪  category-color mappings.
# c(): combines values into a vector, converting them to a common type when needed.
# scale_x_discrete(): controls category order, labels, and spacing on the x-axis.
# scale_y_continuous(): controls numeric y-axis breaks, labels, limits, and spacing.
# seq(): creates an equally spaced sequence; by sets spacing and length.out sets the
 ↪  number of values.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# theme(): customizes non-data plot elements such as legends, grids, and axis text.
# as.data.frame() reshapes the percentage table into one row per quality-price
 ↪  combination.
# position_stack() places segments and labels within each 100% quality-group bar.
bar_data <- as.data.frame(row_pct, responseName = "Percent")
bar_data$Quality <- factor(bar_data$Quality, levels = quality_levels)
bar_data$Price <- factor(bar_data$Price, levels = price_labels)
bar_data$Label <- ifelse(bar_data$Percent >= 8, sprintf("%.1f%%",bar_data$Percent), "")
quality_axis <- paste0(quality_levels, "\n(n = ", rowSums(counts), ")")
ggplot(bar_data, aes(Quality, Percent, fill = Price)) +
  geom_col(position = position_stack(reverse = TRUE), width = .62,
           colour = "white", linewidth = .4) +
  geom_text(aes(label = Label), position = position_stack(vjust = .5, reverse = TRUE),
            colour = "#22313C", size = 3.8) +
  scale_fill_manual(values = c("#DCE8EF", "#ACC9DA", "#E8CFA5", "#C9D8BD"), drop = FALSE)
   ↪  +
  scale_x_discrete(labels = quality_axis) +
  scale_y_continuous(breaks = seq(0,100,25), limits = c(0,100)) +
  labs(title = "Price distributions differ across quality categories",
       x = "Quality rating", y = "Within-group percentage", fill = "Meal price (USD)") +
  theme(legend.position = "bottom")
```

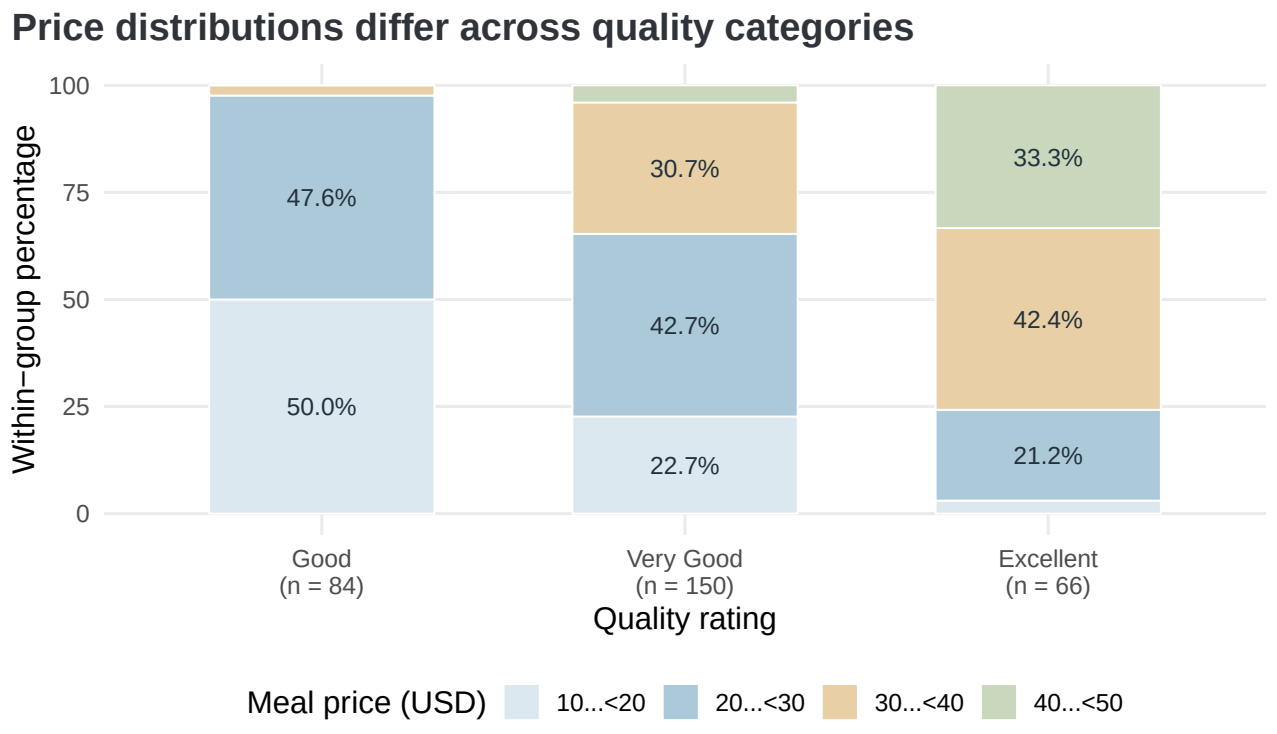


Figure 1: Original R chart from the checked Restaurant CSV. Every bar represents 100% of its quality group. Segment labels are percentages; n gives group size.

Small segments are readable in the table even when too narrow for a label. Middle segments in stacked bars have different baselines, so use the table for precise comparisons.

Column percentages: quality distribution within price

```
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# colSums(): sums each column separately.
# column_pct uses prop.table(counts, margin = 2): divide by each column total.
# rbind() adds a totals row so each price-band distribution can be checked against 100%.
kable(rbind(column_pct, Total = colSums(column_pct)), digits = 1,
      caption = "Within each price band: percentage in each quality category. Each column
       ↪  totals 100% before rounding.")
```

Table 3: Within each price band: percentage in each quality category. Each column totals 100% before rounding.

	10–<20	20–<30	30–<40	40–<50
Good	53.8	33.9	2.6	0.0
Very Good	43.6	54.2	60.5	21.4
Excellent	2.6	11.9	36.8	78.6
Total	100.0	100.0	100.0	100.0

Among the **28 restaurants in the highest band**, **22 are Excellent**, or **78.6%**. Among the **66 Excellent restaurants**, **22 are in the highest band**, or **33.3%**. Reversing the conditioning changes the denominator and the answer.

```
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
# Index counts and percentages by quality row and price-band column.
# stopifnot() stops execution if the count or either conditioning denominator is
 ↪  incorrect.
stopifnot(counts["Very Good",2] == 64,
          abs(row_pct["Very Good",2]-100*64/150) < 1e-10,
          abs(column_pct["Very Good",2]-100*64/118) < 1e-10)
```

i Calculate from counts, then round

Our percentages come directly from the checked counts. Round only for display. Independently rounded cells may not total exactly 100%. The book's Table 2.11 prints some cells differently (for example 33.4% for 22/66); our calculated value rounds to 33.3%. We retain the observations and reproducible arithmetic.

Crosstabs show an association between quality and price bands here. They do not show that raising a price improves quality. Grouping boundaries can affect the display, and combining subgroups can hide or reverse patterns. The book develops this aggregation issue as Simpson's paradox in §2.3. A full subgroup analysis would require relevant subgroup data, which this Restaurant file does not contain.

Business case: commercials and weekly electronics sales

A San Francisco electronics store records **10 weeks** of television-commercial counts and sales. One row pairs the commercials and sales **from the same week**. Sales are recorded in **hundreds of US dollars**: a value of 50 means USD 5,000. The question is: **Do weeks with more commercials tend to have higher sales?** These historical observations can describe an association, but they do not isolate the effect of advertising.

يسجّل متجر إلكترونيات في سان فرانسيسكو عدد الإعلانات التلفزيونية والمبيعات خلال **عشرة أسابيع**. يجمع كل صف عدد الإعلانات والمبيعات **للأسبوع نفسه**. سُجّلت المبيعات **بمئات الدولارات الأمريكية**؛ فالقيمة 50 تعني خمسة آلاف دولار. والسؤال هو: **هل تميل الأسابيع التي تضم إعلانات أكثر إلى تسجيل مبيعات أعلى؟** تصف هذه المشاهدات التاريخية ارتباطًا، لكنها لا تعزل أثر الإعلان في المبيعات.

Download Electronics CSV. The original workbook is named Stereo.xls; all ten observations match the book's Electronics table. `week` identifies the observation, while `commercials` and `sales_hundreds_usd` form our variable pair.

```
# c(): combines values into a vector, converting them to a common type when needed.
# kable() displays each week together with its commercials and sales observations.
# col.names changes the visible headings, including the sales unit, without changing the
 ↪  data.
kable(electronics, col.names = c("Week", "Commercials", "Sales (hundreds of USD)"))
```

Week	Commercials	Sales (hundreds of USD)
1	2	50
2	5	57
3	1	41
4	3	54
5	4	54
6	1	38
7	5	63
8	3	48
9	4	59
10	2	46

Scatterplots: inspect the paired pattern

A **scatterplot** places one numerical variable on each axis and one point for each paired observation. Keep each week's pair together. Sorting the two columns independently destroys the observed relationship.

```
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# scale_y_continuous(): controls numeric y-axis breaks, labels, limits, and spacing.
# expansion(): sets extra axis space; mult is relative padding and add is padding in axis
 ↪  units.
# c(): combines values into a vector, converting them to a common type when needed.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# geom_point() draws each paired week; geom_text() labels points with week identifiers.
# nudge_y moves labels above points, and scale_x_continuous() sets the count-axis tick
 ↪  marks.
ggplot(electronics, aes(commercials, sales_hundreds_usd)) +
  geom_point(size = 3.2, colour = "#2878A5") +
  geom_text(aes(label = week), nudge_y = 1, size = 3.5, colour = "#56616D") +
  scale_x_continuous(breaks = 1:5) +
  scale_y_continuous(expand = expansion(mult = c(.06,.12))) +
  labs(title = "More commercials accompany higher weekly sales",
       x = "Number of television commercials", y = "Weekly sales (hundreds of USD)")
```

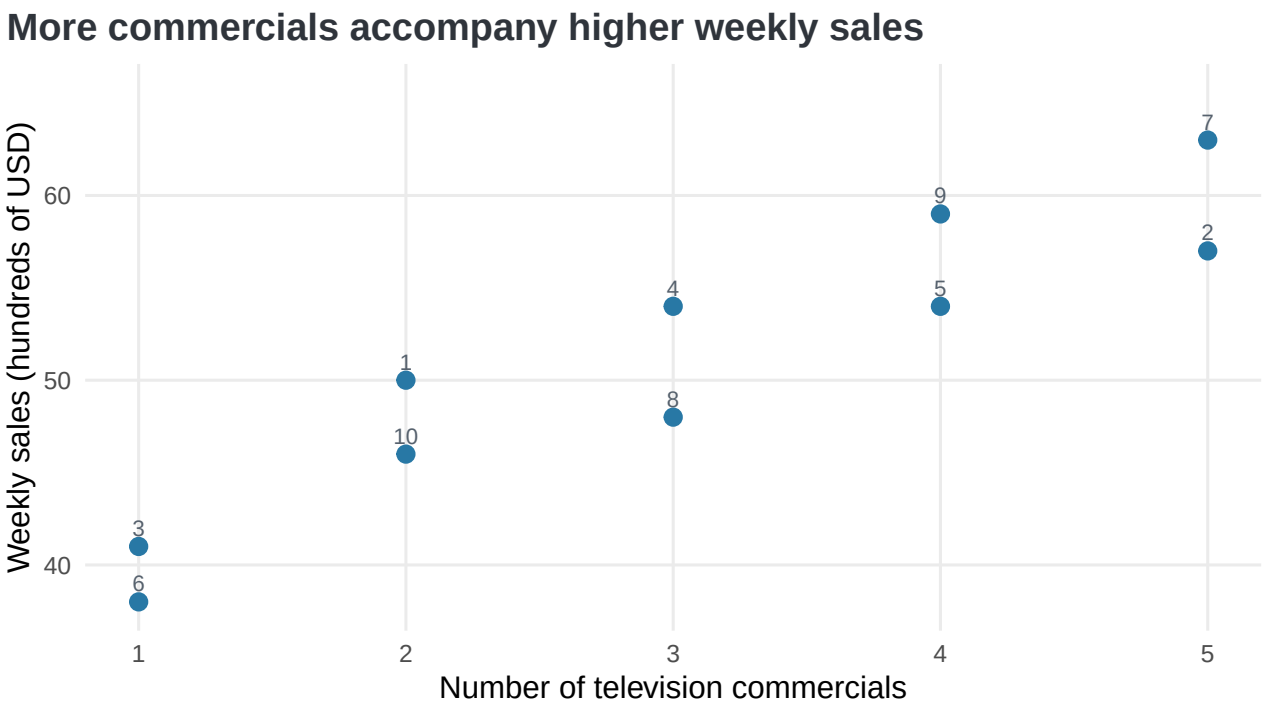


Figure 2: Original R scatterplot of the ten checked Electronics observations. Labels identify weeks. No fitted regression model is required for this descriptive display.

Read four features before calculating a coefficient:

- **Direction:** as commercials increase, sales tend to increase. This is a positive association.
- **Form:** the pattern is approximately straight rather than clearly curved.
- **Strength:** points cluster fairly closely around an upward linear pattern, with variation at each commercial count.
- **Unusual observations:** look for isolated points and values far from the overall pattern. The plot alone cannot establish whether a record is erroneous.

The count is on the horizontal axis because it is the proposed explanatory variable for this question. That placement does not establish cause and effect. We leave connecting lines out because our immediate question concerns the paired relationship, rather than a chronological sales path.

Covariance: do deviations move together?

The **sample covariance** is

$$s_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n - 1}.$$

When both observations are above their means, or both below, their product of deviations is positive. Opposite signs give a negative product. Summing these products describes whether deviations tend to move together.

```
# data.frame(): builds a table whose named columns have compatible lengths.
# mean(): computes an average; logical TRUE/FALSE values contribute 1/0, giving a
 ↪  proportion.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# Subtract each sample mean and multiply the paired deviations week by week.
```

```r
# sum() combines the cross-products; cov() divides their sum by n-1 for sample covariance.
x <- electronics$commercials
y <- electronics$sales_hundreds_usd
products <- data.frame(Week = electronics$week, x = x, y = y,
                       x_deviation = x-mean(x), y_deviation = y-mean(y),
                       Product = (x-mean(x))*(y-mean(y)))
kable(products, digits = 1)
```

Week	x	y	x_deviation	y_deviation	Product
1	2	50	-1	-1	1
2	5	57	2	6	12
3	1	41	-2	-10	20
4	3	54	0	3	0
5	4	54	1	3	3
6	1	38	-2	-13	26
7	5	63	2	12	24
8	3	48	0	-3	0
9	4	59	1	8	8
10	2	46	-1	-5	5

```r
stopifnot(mean(x) == 3, mean(y) == 51, sum(products$Product) == 99,
          cov(x,y) == 11)
```

Here $\bar{x} = 3$, $\bar{y} = 51$, and the sum of products is 99. Thus $s_{xy} = 99/9 = 11$, in **commercials × hundreds of USD**. Positive covariance agrees with the upward scatterplot.

Covariance depends on measurement units. Its sign helps describe direction, but its magnitude alone cannot establish that an association is strong. For a complete population, the population covariance uses population means and denominator N. R's `cov()` uses the sample denominator $n - 1$.

Correlation: standardize the linear association

The **Pearson sample correlation** divides covariance by the sample standard deviations:

$$r = \frac{s_{xy}}{s_x s_y}.$$

It requires paired numerical observations with nonzero variation in both variables. It is **unitless**, ranges from **−1 to +1**, and describes **linear** association:

Value	Meaning
$r = +1$	All points lie on a straight line with positive slope
$r = -1$	All points lie on a straight line with negative slope
r near zero	Little linear association; a nonlinear relationship may remain
Larger $\lvert r \rvert$	Stronger linear association, interpreted with the plot and context

```r
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
# cor() calculates Pearson linear correlation from the paired observations.
# Dividing cov(x,y) by sd(x)*sd(y) checks the same unitless calculation explicitly.
r <- cor(x,y)
kable(data.frame(Quantity = c("Sample SD: commercials", "Sample SD: sales (hundreds of
 ↪  USD)",
                              "Sample covariance", "Pearson correlation"),
                 Value = c(sd(x),sd(y),cov(x,y),r)), digits = 4)
```

Quantity	Value
Sample SD: commercials	1.4907
Sample SD: sales (hundreds of USD)	7.9303
Sample covariance	11.0000
Pearson correlation	0.9305

```r
stopifnot(abs(r-cov(x,y)/(sd(x)*sd(y))) < 1e-10,
          abs(r-0.9304906) < 1e-7)
```

Interpretation: across these ten weeks, commercials and sales have a strong positive linear association, $r \approx 0.9305$, consistent with the book's rounded value 0.93. This does not mean "sales increase by 93%," and correlation does not give a change in sales per additional commercial. Estimating such a change requires a model, developed later in regression.

Changing units: covariance changes, correlation stays

```r
# data.frame(): builds a table whose named columns have compatible lengths.
# c(): combines values into a vector, converting them to a common type when needed.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
# Multiply recorded sales by 100 to convert hundreds of USD into USD.
# cov() scales by the same factor; cor() is unchanged under this positive unit conversion.
y_usd <- 100*y
units <- data.frame(Sales_unit = c("Hundreds of USD", "USD"),
                    Sample_covariance = c(cov(x,y),cov(x,y_usd)),
                    Correlation = c(cor(x,y),cor(x,y_usd)))
kable(units, digits = 4)
```

Sales_unit	Sample_covariance	Correlation
Hundreds of USD	11	0.9305
USD	1100	0.9305

```r
stopifnot(abs(cov(x,y_usd)-100*cov(x,y)) < 1e-10,
          abs(cor(x,y_usd)-cor(x,y)) < 1e-10)
```

Converting sales to USD changes covariance from **11 to 1100**, but the correlation stays **0.9305**. The underlying relationship has not changed. Positive changes of measurement units and shifts of origin leave Pearson correlation unchanged. Correlation also treats x and y symmetrically: `cor(x,y)` equals `cor(y,x)`.

Why a correlation coefficient needs its plot

The following are **artificial teaching patterns**, not Electronics data. The curved pattern uses $y = x^2$ for $x = -4, -3, \ldots, 4$.

```
# data.frame(): builds a table whose named columns have compatible lengths.
# factor(): stores categorical values; levels fixes their permitted order and labels can
 ↪  rename their display.
# c(): combines values into a vector, converting them to a common type when needed.
# stopifnot(): stops execution if any supplied condition is FALSE; successful checks
 ↪  produce no output.
# abs(): returns absolute values; a small tolerance in an equality check permits
 ↪  floating-point rounding.
# ggplot(): starts a plot from data; additional layers are joined with +.
# aes(): maps variables to visual properties such as x, y, fill, color, or group.
# geom_point(): draws a point for each row at its mapped x and y values.
# labs(): sets axis titles, legend titles, plot titles, and captions.
# rbind() combines three explicitly artificial paired patterns: y=x, y=-x, and y=x^2.
# facet_wrap() separates the plots; cor() verifies that a curve can have zero linear
 ↪  correlation.
u <- -4:4
patterns <- rbind(data.frame(Pattern = "Positive line: r = 1", x = u, y = u),
                  data.frame(Pattern = "Negative line: r = -1", x = u, y = -u),
                  data.frame(Pattern = "Curve: r = 0", x = u, y = u^2))
patterns$Pattern <- factor(patterns$Pattern,
  levels = c("Positive line: r = 1", "Negative line: r = -1", "Curve: r = 0"))
stopifnot(cor(u,u) == 1, cor(u,-u) == -1, abs(cor(u,u^2)) < 1e-10)
ggplot(patterns, aes(x,y)) +
  geom_point(size = 2.5, colour = "#2878A5") +
  facet_wrap(~Pattern, nrow = 1, scales = "free_y") +
  labs(title = "Zero correlation can conceal a clear relationship",
       x = "x (arbitrary units)", y = "y (arbitrary units)")
```

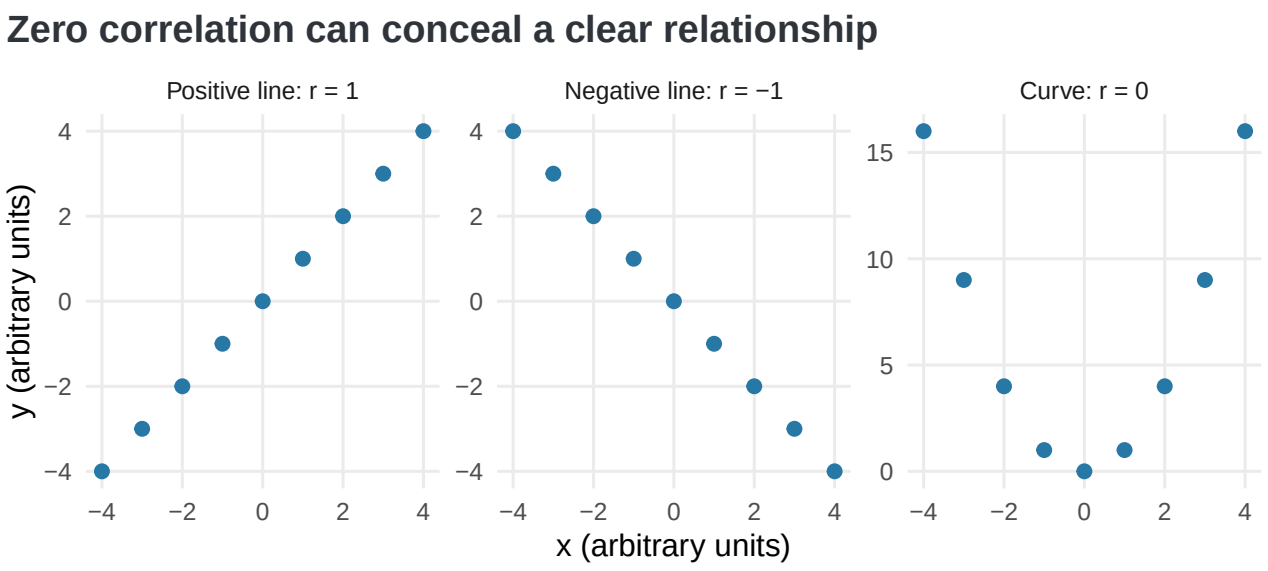


Figure 3: Original artificial examples. Pearson correlation captures straight-line direction but can be zero despite a clear curved relationship. Panel axes use arbitrary units.

Keep these limitations in view:

- **Nonlinearity:** $r = 0$ does not establish that variables are unrelated or independent.
- **Influential observations:** an isolated observation can substantially change correlation. Investigate it and compare results transparently; preserve valid records.
- **Restricted range and mixed groups:** a narrow observed range or a mixture of subgroups can change the apparent association. Inspect context and subgroup information when available.
- **Causation:** seasonal demand, promotions, or other unmeasured factors might affect both advertising and sales. These are possible explanations to investigate, not recorded facts about this store.
- **Scope:** ten observed weeks do not establish a stable relationship for all future weeks. Repeated weeks may also be dependent; calculation alone does not justify inferential claims.

Computing Pearson correlation requires no normality assumption by itself. Hypothesis tests and confidence intervals for association require additional assumptions and belong to later modules.

Summary: a defensible relationship description

Task	What to report	Essential check
Compare categorical distributions	Counts, group sizes, and conditional percentages	State the conditioning group and denominator
Compare numerical distributions by group	Group centers, spreads, and aligned plots	Use meaningful units and the same conventions
Describe two numerical variables	Scatterplot and, when suitable, covariance and correlation	Keep pairs intact; check shape and influential observations
Explain the result	An association in the observed context	Separate descriptive evidence from causal claims

For Restaurant, conditional price distributions differ across ratings. For Electronics, weekly commercials and sales show a strong positive linear association. Neither result establishes a causal effect.

Complete Lesson 4 practice, then complete the integrated module review.

Sources and data

- Anderson et al. (2020), Essentials of Modern Business Statistics with Microsoft Excel, 8th ed.: §2.3, pp.65–69, Tables 2.9–2.11; §2.4, pp.75–80, Table 2.14; §3.5, pp.144–152, Table 3.6 and the covariance/correlation example.
- Restaurant CSV: first ten displayed records and all twelve cells of Table 2.10 match; remaining individual records are not printed for comparison.
- Electronics CSV: all ten paired observations match Table 2.14 and the repeated example in Table 3.6. R calculations reproduce covariance 11 and correlation rounded to 0.93.
- All charts and numerical tables here are original R outputs from these CSVs, except the explicitly artificial correlation-pattern illustration. Data guide.